

**S HECATE STRAIT BUOY 46185, Canada**WMO: **996530**Lat: **52.420N**Lon: **129.790W**Elev: **10**StdP: **14.69**Time Zone: **-9.00 (W09)**Period: **01-18**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 2             | 28.2       | 32.1 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 42.4                  | 44.8 | 40.1 | 45.6 | 19.1                     | 20   | 0.816 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 8             | 2.9                    | 63.0            | N/A  | 61.7 | N/A  | 60.7 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 9.2                  | 310  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 40.7          | 36.2 | 32.2 | DB                         | 26.8 | 65.0               | 5.5 | 4.6                                                | 22.8 | 68.3       | 19.6 | 71.0       | 16.5 | 73.6       | 12.5 | 76.9 | (4) |
| (5)               |      |      | WB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  | N/A  | (5) |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 49.2   | 43.7  | 42.9  | 43.0  | 45.3  | 48.8  | 53.1  | 56.5  | 58.3  | 56.0  | 51.7  | 46.7  | 44.1 |
|                                                                                  | DBStd       | 6.35   | 4.52  | 3.87  | 3.92  | 2.46  | 2.27  | 2.60  | 2.33  | 2.41  | 1.87  | 2.55  | 4.19  | 4.12 |
|                                                                                  | HDD50       | 1107   | 196   | 198   | 216   | 141   | 49    | 3     | 0     | 0     | 0     | 13    | 106   | 185  |
|                                                                                  | HDD65       | 5761   | 661   | 618   | 681   | 590   | 502   | 356   | 265   | 209   | 269   | 412   | 549   | 649  |
|                                                                                  | CDD50       | 823    | 0     | 0     | 0     | 1     | 12    | 97    | 200   | 257   | 181   | 66    | 7     | 2    |
|                                                                                  | CDD65       | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 16.1   | 19.9  | 17.8  | 18.1  | 15.4  | 13.0  | 12.9  | 12.8  | 12.3  | 14.3  | 17.0  | 19.4  | 20.1 |
| Precipitation                                                                    | PrecAvg     | 116.7  | 13.8  | 9.8   | 10.6  | 8.3   | 5.5   | 4.7   | 4.0   | 5.3   | 8.1   | 14.8  | 16.7  | 15.2 |
|                                                                                  | PrecMax     | 145.8  | 20.7  | 16.9  | 20.6  | 16.0  | 11.2  | 8.1   | 11.5  | 11.2  | 15.0  | 23.8  | 23.9  | 25.7 |
|                                                                                  | PrecMin     | 93.9   | 6.7   | 4.3   | 4.0   | 3.4   | 1.8   | 1.7   | 0.9   | 1.3   | 1.3   | 6.7   | 9.7   | 6.3  |
|                                                                                  | PrecStd     | 13.8   | 3.5   | 3.3   | 3.1   | 2.2   | 2.0   | 1.5   | 2.4   | 2.2   | 3.3   | 4.4   | 3.9   | 4.1  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 51.2  | 50.4  | 50.4  | 51.5  | 54.5  | 62.1  | 62.7  | 65.6  | 61.9  | 57.9  | 54.7  | 53.9 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | DB     | 50.0  | 49.3  | 49.3  | 50.1  | 53.1  | 59.5  | 61.5  | 64.2  | 60.4  | 56.7  | 53.0  | 51.6 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | DB     | 49.1  | 48.3  | 48.2  | 49.2  | 52.3  | 58.0  | 60.6  | 62.8  | 59.5  | 55.8  | 52.2  | 50.0 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | DB     | 48.3  | 47.2  | 47.1  | 48.4  | 51.7  | 56.7  | 59.8  | 61.4  | 58.7  | 54.9  | 51.3  | 48.9 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 3.9   | 3.7   | 3.9   | 3.3   | 2.7   | 2.6   | 2.7   | 2.9   | 2.7   | 3.1   | 4.0   | 4.1  |
|                                                                                  |             | MCDBR  | 3.9   | 3.6   | 3.9   | 3.6   | 3.3   | 3.7   | 3.4   | 4.0   | 3.1   | 3.5   | 4.0   | 4.2  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.317  | 0.323 | 0.349 | 0.385 | 0.383 | 0.374 | 0.370 | 0.373 | 0.364 | 0.347 | 0.324 | 0.309 |      |
|                                                                                  | taud        | 2.414  | 2.497 | 2.434 | 2.313 | 2.381 | 2.448 | 2.512 | 2.505 | 2.500 | 2.461 | 2.377 | 2.358 |      |
|                                                                                  | Ebn at Noon | 216    | 254   | 268   | 269   | 276   | 279   | 278   | 271   | 260   | 239   | 208   | 194   |      |
|                                                                                  | Edh at Noon | 18     | 23    | 29    | 36    | 36    | 34    | 31    | 30    | 27    | 23    | 19    | 16    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 192    | 465   | 780   | 1237  | 1599  | 1675  | 1619  | 1381  | 949   | 524   | 246   | 161   |      |
|                                                                                  | RadStd      | 26     | 83    | 106   | 79    | 156   | 151   | 195   | 145   | 103   | 77    | 34    | 25    |      |

**Historical Trends**

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (15 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page